FINAL PROJECT REPORT · AI-ML-210

Ensemble Methods: Boosting vs. Bagging

An empirical, from-scratch comparison of AdaBoost and Random Forest on tabular data

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The Central Research Question

"Under what conditions does boosting outperform bagging, and vice versa, and why?"

1 **From-scratch implementation**

Decision Tree, AdaBoost, and Random Forest built directly in Python/NumPy.

2 **Controlled experiments**

Three real-world datasets spanning different sizes, dimensionality and class balance.

3 **Multiple lenses**

Scaling behaviour, noise robustness, and bias–variance trade-offs are all examined.

4 **Unsupervised complement**

PCA is used to explore data structure and class separability before classification.

Foundations from the Literature

A	**Decision Trees**	Recursively partition the feature space to minimise impurity; CART uses Gini impurity or entropy as the split criterion.
B	**Boosting (AdaBoost)**	Iteratively re-weights samples, giving higher importance to misclassified instances each round.
C	**Bagging & Random Forests**	Bagging trains bootstrap replicates to reduce variance; Random Forests add feature sub-sampling to decorrelate trees further.
D	**Unsupervised Techniques**	PCA finds directions of maximum variance; K-Means partitions data into k clusters; DBSCAN discovers density-based clusters of arbitrary shape.

Decision Tree (CART)

- Follows the CART methodology with exhaustive split search
- Supports both Gini impurity and entropy as split criteria
- sample_weight parameter is supported to serve the boosting module
- Split criterion maximises impurity reduction ΔI at every node

Impurity Reduction

$$\Delta I = I(\text{parent}) - (N\_\text{left}/N)\cdot I(\text{left}) - (N\_\text{right}/N)\cdot I(\text{right})$$

Gini impurity:

$$I\_G(p) = 1 - \Sigma\text{Ʒ}\ p\text{Ʒ}^2$$

Entropy:

$$I\_E(p) = -\ \Sigma\text{Ʒ}\ p\text{Ʒ}\cdot\log_2(p\text{Ʒ} + \varepsilon)$$

AdaBoost (Discrete SAMME)

- Uses depth-1 decision stumps as weak learners
- Sample weights are updated after every boosting round
- $\alpha_m = \ln((1 - \varepsilon_m)/\varepsilon_m)$ for the binary classification case
- Final prediction is a weighted majority vote across all stumps

Weight Update Rule

$$w_i^{(m+1)} = w_i^{(m)} \cdot \exp(\alpha_m \cdot \mathbb{1}[h_m(x_i) \neq y_i])$$

Amplifier term:

$$\alpha_m = \ln((1-\varepsilon_m) / \varepsilon_m)$$

Misclassified samples receive exponentially larger weight, forcing the next stump to focus on the hardest cases.

Random Forest & Unsupervised Pipeline

Random Forest

- Bootstrap samples are drawn with replacement
- Trees grown with feature sub-sampling (sqrt by default)
- Out-of-bag (OOB) error computed per sample using the trees where it was never included

Unsupervised Pipeline

- PCA via eigen-decomposition of the covariance matrix
- K-Means with Lloyd’s algorithm, using multiple restarts
- DBSCAN using brute-force region queries
- Adjusted Rand Index (ARI) reported against true labels where applicable

Three Datasets, Three Profiles

Dataset	Samples	Features	Classes
Adult Income	48,842	14	2 (imbalanced)
Covertype (subset)	5,000	54	7 (imbalanced)
WDBC	569	30	2

Adult Income

Large-scale binary classification benchmark with strong class imbalance — the stress test for scalability.

Covertype

Multi-class (7-way), high-dimensional subset used to probe how ensembles cope with complexity.

WDBC

Small, clean, low-dimensional dataset used as the reference case for a low-noise regime.

Preprocessing, Metrics & Experiment Plan

Preprocessing

- Features standardised to zero mean, unit variance
- No missing values present in any dataset
- Random oversampling (imbalanced-learn) applied to Adult and Covertype minority classes

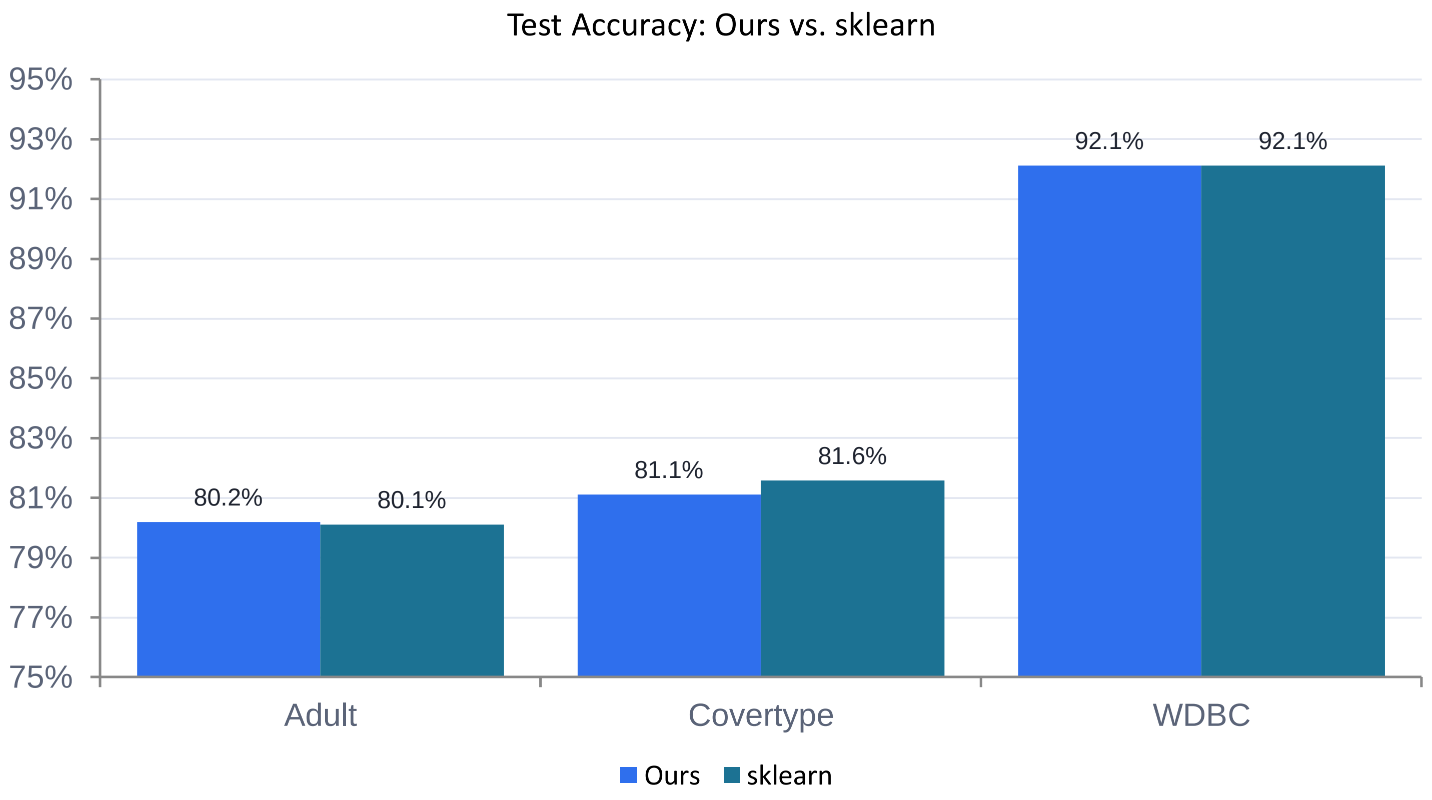
Metrics & Protocol

- Accuracy, macro F1, and AUC-ROC (micro-averaged for multi-class)
- 5-fold cross-validation with a fixed random seed (42)

Seven Experiments

1. Baseline validation — single tree vs. sklearn
2. AdaBoost scaling — n_estimators 1→200
3. Random Forest scaling — n_estimators & max_depth
4. Head-to-head — 100 estimators, 5-fold CV
5. Noise robustness — 5%, 10%, 20% label noise
6. Bias–variance decomposition — 100 replicates on WDBC
7. Unsupervised analysis — PCA, K-Means, DBSCAN

Baseline Validation: Our Tree vs. scikit-learn

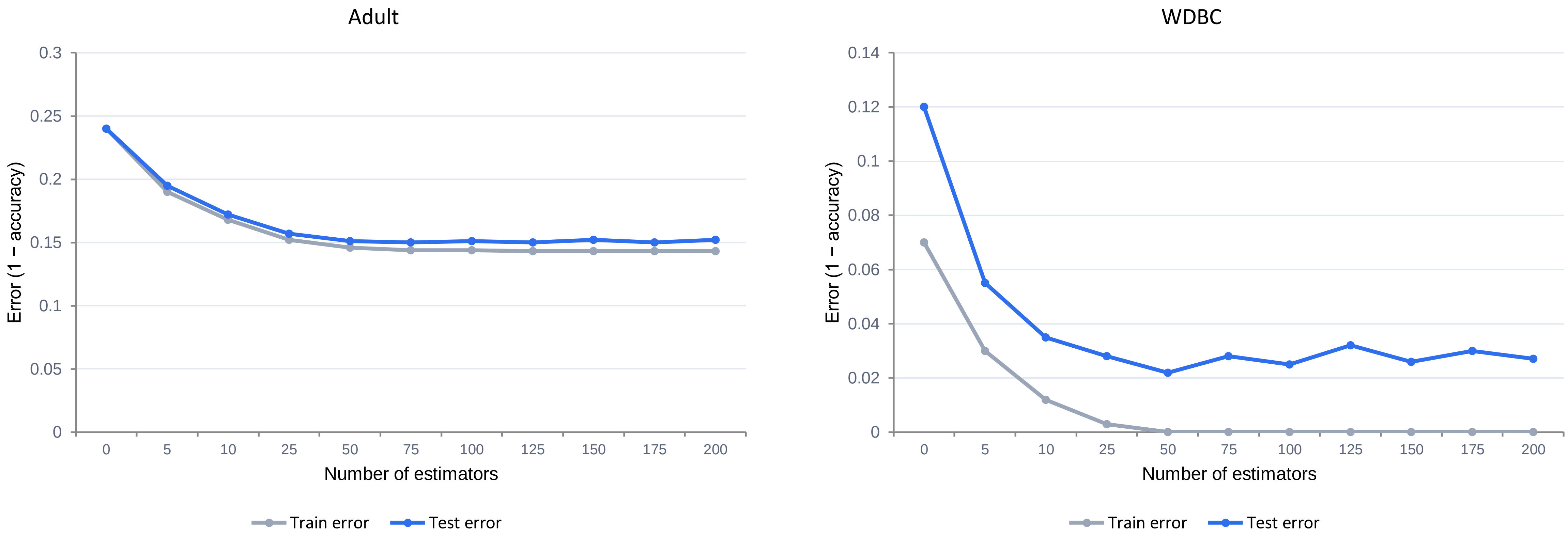


Within 2%

agreement with scikit-learn across accuracy, macro F1, and AUC-ROC on all three datasets

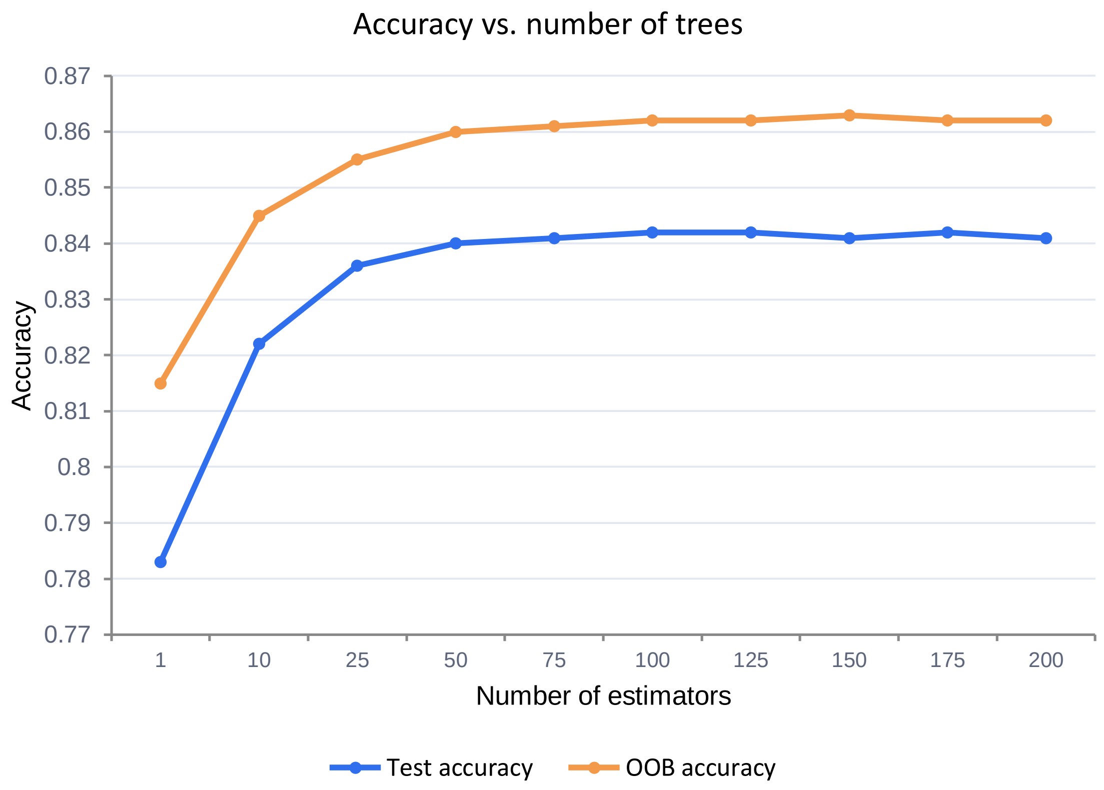
Confirms the from-scratch Decision Tree satisfies the project's correctness requirement before it is reused inside AdaBoost and Random Forest.

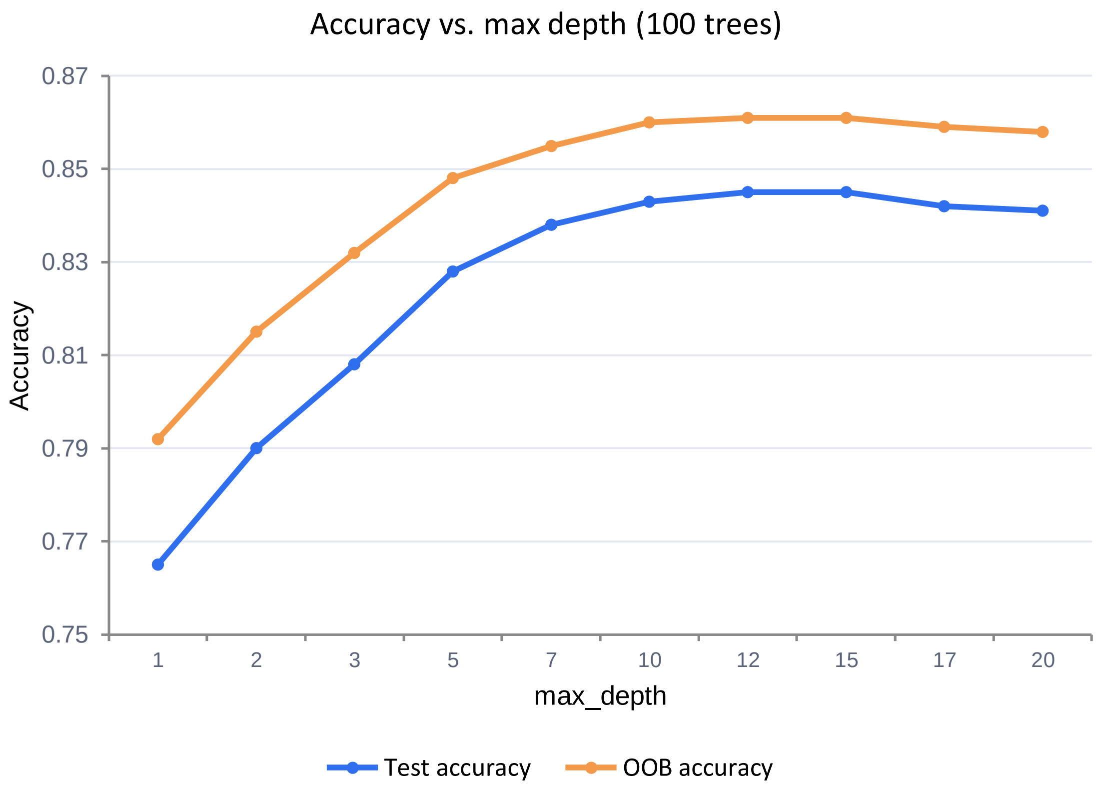
AdaBoost Scaling: Error vs. Number of Stumps



Accuracy improves rapidly to about 50 estimators, then plateaus — diminishing returns with no pronounced overfitting, consistent with boosting theory.

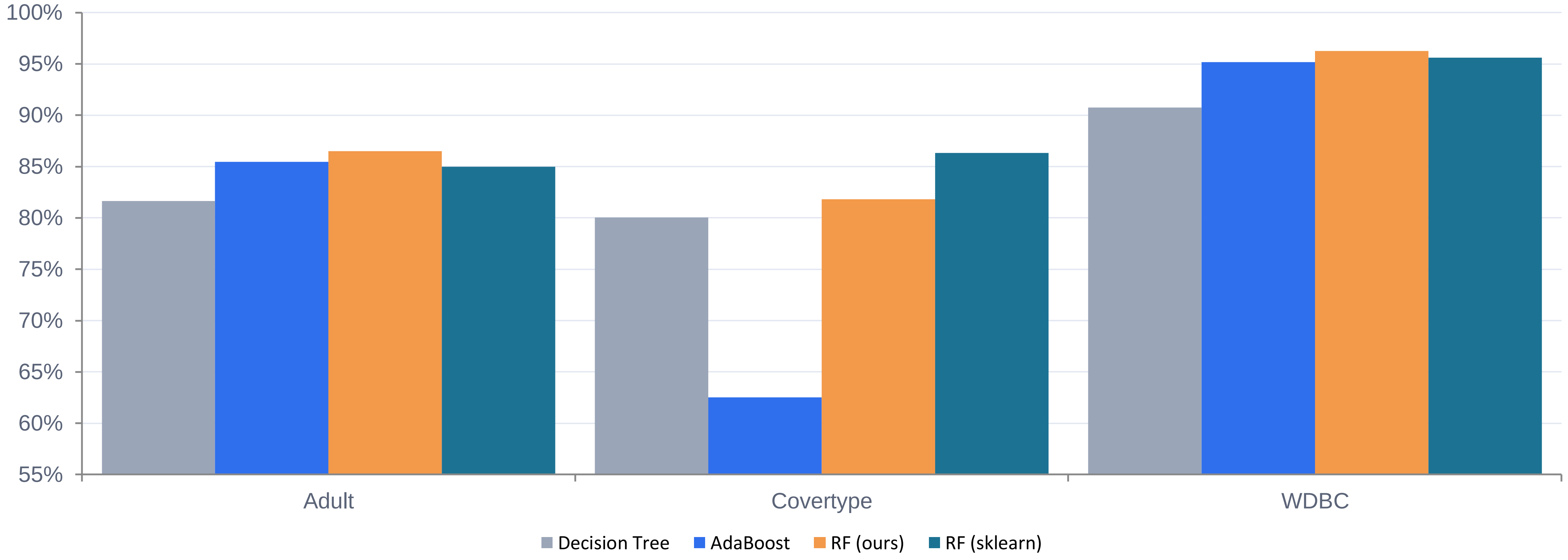
Random Forest Scaling on Adult





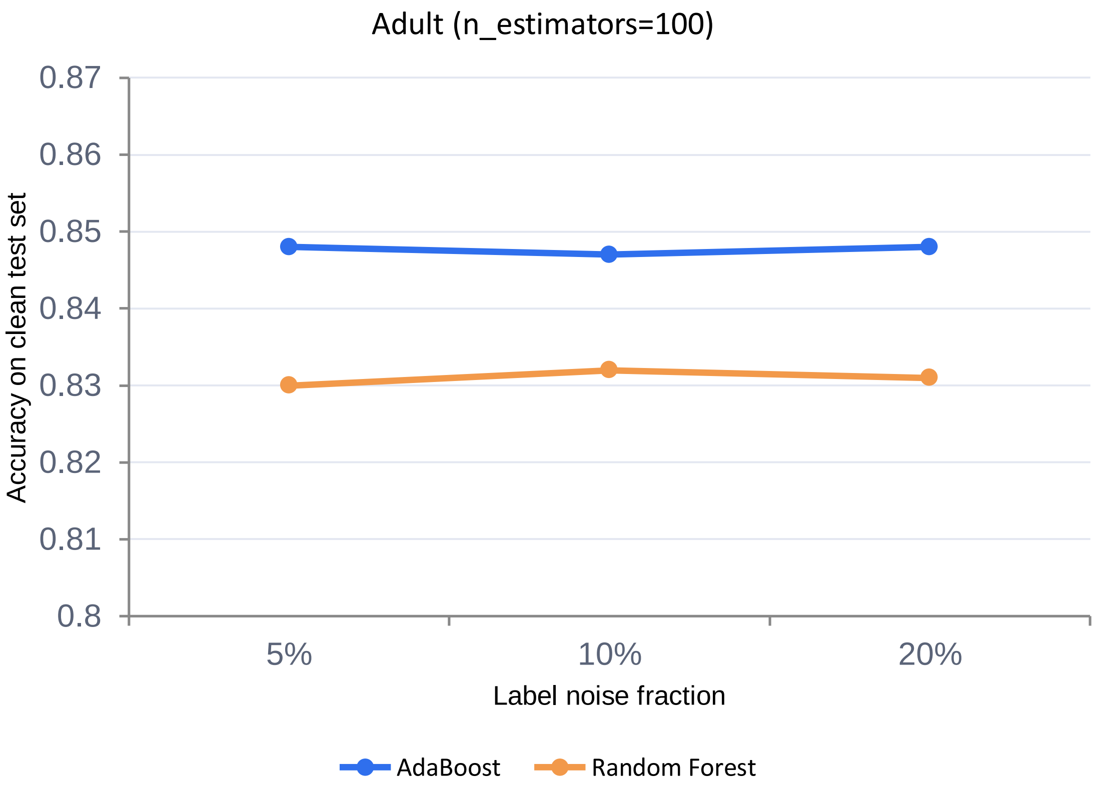
Both curves stabilise around 100 trees, with OOB accuracy giving a reliable estimate of test performance. Optimal depth for this dataset sits around 10–15.

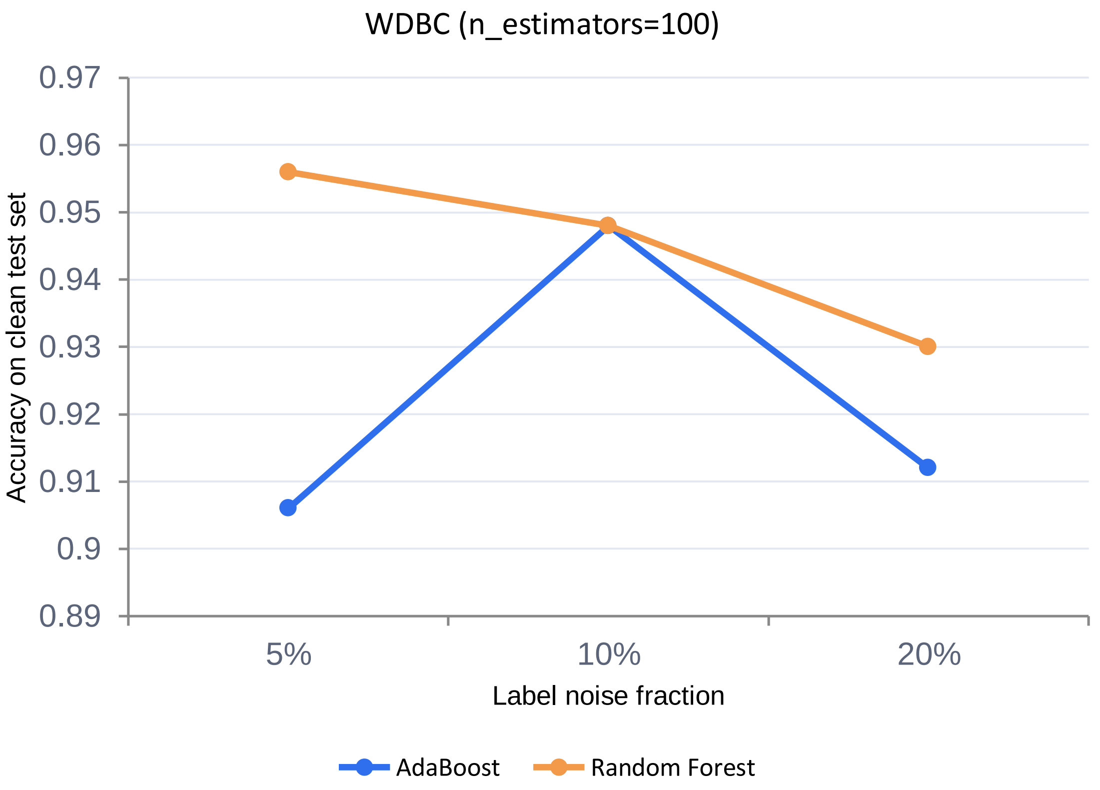
Head-to-Head: Accuracy Across Models & Datasets



Random Forest leads on Adult and Covertype; AdaBoost achieves the highest accuracy on the clean, low-noise WDBC dataset. Ensembles beat the single tree everywhere.

Noise Robustness: Accuracy Under Label Noise





AdaBoost suffers a sharper drop under high noise, confirming its sensitivity to mislabeled instances. Random Forest maintains higher accuracy, especially on WDBC.

Bias–Variance Decomposition & Unsupervised View

Bias–Variance on WDBC (100 bootstrap replicates)

Model	Bias²	Variance	Brier Score
AdaBoost	0.0179	0.0080	0.0258
Random Forest	0.0294	0.0038	0.0332

AdaBoost exhibits lower bias but higher variance; boosting reduces bias at the cost of variance, while bagging reduces variance.

Unsupervised Analysis (PCA)

7 PCs retain 90.5% of variance on WDBC

16 PCs retain 92.8% of variance on Adult

ARI < 0.5 K-Means / DBSCAN clusters do not align well with true classes

Class separability in the low-dimensional PCA projections is moderate, with some overlap — explaining why ensembles still show room for improvement.

Boosting Reduces Bias, Bagging Reduces Variance

Boosting outperforms bagging when data is clean and the weak learners are sufficiently weak; bagging is preferred when noise is present or data is high-dimensional and complex.

On clean data (WDBC)

AdaBoost achieves higher accuracy, minimising bias by focusing on hard examples

Under label noise

AdaBoost's aggressive re-weighting overfits noisy points; Random Forest is more robust

On large/complex data

Random Forest outperforms AdaBoost on Adult and Covertype thanks to bootstrap aggregation and feature sub-sampling

Future work: Gradient Boosting Machines (GBM), t-SNE for deeper unsupervised structure, systematic hyperparameter tuning, and evaluation on larger datasets.